

BOAMP Digital Procurement Study

Successor Linkage, Survival, and Trend Methodology

BOAMP Data Science Internship Project

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Technical Summary

This report states the current defensible BOAMP successor-linkage pipeline. The study does not claim to certify legal contract renewals. Its event is an *observable successor procurement*: a later BOAMP procurement episode from the same buyer that is sufficiently similar to an earlier awarded digital procurement episode.

The current Grand Ouest survival cohort contains 3,800 awarded digital procurement episodes. The main linkage rule accepts 544 successor links, giving an observed event rate of 0.143 and a median observed successor time of 31.82 months among linked events. The candidate pool is broad: 763,417 candidate pairs from 3,520 anchors with at least one candidate.

The current national development reference contains 252 bootstrap-labelled anchors and 7,031 labelled anchor-candidate rows. On its held-out internal split, `M_B_text_ranking @ 0.70` has precision 0.800, recall 0.182, false-positive rate 0.000, and 5 accepted links. These are internal protocol-reference estimates, not independently validated accuracy estimates, because both label passes were generated by the deterministic bootstrap rules in `scripts/auto_annotate_wavela.py`. `M_C_weighted_gated` recovers more positives (recall 0.409) but admits more false positives (FPR 0.105). This trade-off is important because a false positive in a survival dataset creates both a false event and a false event time, while an abstained link is handled conservatively as right-censoring.

1 Context

Gigalis is the digital central purchasing body (*centrale d'achat numérique*) for public institutions in the Pays de la Loire region and beyond. It designs pooled framework agreements in cloud, cybersecurity, digital infrastructure, and artificial intelligence, and its strategic relevance depends on anticipating members' future needs. The internship's central question is: using historical public digital procurement data, how can the probability that a contract or technological segment generates an identifiable purchasing need within the next 12–24 months be estimated, and how can that estimate inform Gigalis's central purchasing strategy?

This decomposes into three sub-problems: a *lifetime problem* (survival analysis: how long between a digital procurement episode and its successor?), a *trend problem* (change-point detection: which technological segments are growing, plateauing, or declining?), and a *text-signal problem* (NLP: extracting a contract's technological theme automatically). This project answers the first two in depth; the NLP sub-problem is addressed with a reproducible coarse substitute rather than a trained classifier (§8).

BOAMP does not encode legal contract renewal as an explicit field, so the study cannot certify legal renewals. It instead constructs and evaluates a proxy: an *observable successor procurement*, a later BOAMP episode from the same buyer that is sufficiently similar to, and temporally plausible relative to, an earlier awarded digital procurement episode. Every downstream result – survival curves, hazard ratios, trend segments – is conditioned on this proxy, not on certified legal ground truth. The strongest defensible framing is: because legal renewal status is not directly observed, the study measures observable successor procurements under a precision-first linkage rule, then assesses how sensitive its descriptive conclusions are to that rule.

2 What The Pipeline Measures

Let i index an earlier awarded procurement episode and j index a later candidate episode. The object of interest is not a legal renewal certificate, because BOAMP notices do not provide that ground truth consistently. Instead, the pipeline estimates whether the data show a later

procurement episode that is credible enough to be treated as a successor:

$$Y_i = \begin{cases} 1, & \text{if one accepted observable successor is found before the cutoff,} \\ 0, & \text{otherwise.} \end{cases}$$

The event time is

$$\tau_i = C_{\hat{j}} - A_i,$$

where A_i is the award-date origin of episode i , $C_{\hat{j}}$ is the first-publication date of the accepted successor, and $\hat{j}$ is the selected candidate. If no accepted successor is found before 2025-12-31, the episode is right-censored.

3 End-to-End Pipeline

The implemented pipeline is:

BOAMP notices $\rightarrow$ standardised notices $\rightarrow$ procurement episodes
 $\rightarrow$ candidate pairs $\rightarrow$ linkage scoring $\rightarrow$ one successor or abstention $\rightarrow$ survival dataset.

Standardisation and feature engineering. The preprocessing stage standardises dates, buyer identifiers, buyer names, CPV codes, text fields, procedure information, framework flags, and duration fields where they are explicitly available. It does not impute missing duration or invent expected expiry dates. Buyer standardisation is deliberately conservative: municipal variants such as commune/ville/mairie can be harmonised, but intercommunal legal forms are preserved as distinct entities, and conflicting validated SIRENs block a buyer match. The current legal-form audit reports 0 accepted links with conflicting validated SIREN evidence and 0 accepted municipal/intercommunal mixes.

Episode reconstruction. BOAMP is notice-level data, but the analysis needs procurement-level events. Therefore notices belonging to the same procurement are reconstructed into an episode. This avoids treating a consultation notice, correction notice, award notice, or lot-level administrative notice as separate renewals.

Candidate generation. Candidate generation is intentionally broad and is not the final model. For an anchor episode with award date A_i and candidate publication date C_j , the candidate is exposed only if

$$A_i + 90 \leq C_j \leq A_i + 2920.$$

The lower bound removes very near follow-up notices and parallel administrative activity. The upper bound is approximately eight years; it keeps the candidate pool inclusive enough for long public-procurement cycles. Precision is then controlled by the selection stage rather than by a narrow time window. The current blocking rule generated 763,417 candidate pairs, with a median of 89 candidates per anchor.

4 Current National Development Reference

The current reference is national rather than Grand Ouest-only. It samples awarded digital procurement episodes across French region groups and CPV divisions. The dev split has 134 anchors and 3,638 pair rows; the validation split has 60 anchors and 1,763 pair rows. Only the probability frame supports national weighted estimates; enrichment frames are used for stress cases and diagnostics.

The annotation caveat is decisive: labels were produced by deterministic bootstrap rules. Pass A and pass B are re-presentations of the same rules, so their agreement measures repeatability rather than independent annotator agreement. The reference is useful for development, stress testing, and split discipline, but it is not official legal renewal truth or independent specialist ground truth.

5 Linkage Algorithms

All methods operate on the same exposed candidate set. This is crucial: the primary method is not comparing text over the whole BOAMP universe. Buyer and time plausibility are imposed before text ranking.

M_A : deterministic evidence. Define B_{ij} as buyer-identity support, P_{ij} as CPV continuity, and T_{ij} as text similarity. The deterministic rule accepts a link only when strong buyer evidence is present, CPV continuity is positive, and a minimum text signal is present:

$$B_{ij} = 1, \quad P_{ij} > 0, \quad T_{ij} \geq t_A.$$

It is interpretable, but it loses recall when CPV or buyer identifiers are missing or noisy.

M_B : text ranking. For each episode, the text fields are converted to TF-IDF vectors x_i and x_j . Text similarity is cosine similarity:

$$T_{ij} = \cos(x_i, x_j) = \frac{x_i \cdot x_j}{\|x_i\| \|x_j\|}.$$

Within the same-buyer, plausible-time candidate set, the method selects

$$\hat{j}_i = \arg \max_j T_{ij},$$

and accepts the selected candidate only if

$$T_{i\hat{j}_i} \geq 0.70.$$

This is the primary method because it is simple, reproducible, auditable, and best matches the precision-first objective.

M_C : weighted gated score. This method combines evidence components into a score:

$$S_{ij} = 0.50B_{ij} + 0.25T_{ij} + 0.20P_{ij} + 0.05G_{ij},$$

where G_{ij} is a timing-plausibility score. Missing components are handled by the implemented score-normalisation logic rather than imputation. The method then applies gates and ranks by S_{ij} . It is useful as a higher-recall contrast, but the validation results show a materially higher false-positive rate.

M_D : Fellegi–Sunter probabilistic linkage. Let γ_{ij} be the discretised comparison vector for buyer, text, CPV, and timing evidence. Fellegi–Sunter estimates the likelihood of observing γ_{ij} under a match class M and a non-match class U , then uses the log-likelihood ratio

$$w_{ij} = \log \frac{P(\gamma_{ij} | M)}{P(\gamma_{ij} | U)}.$$

The fitted model provides `fs_match_weight` and `fs_match_probability`. In the current data it does not outperform the simple text-ranking rule, likely because true successors are rare and same-buyer procurement activity contains many non-renewal lookalikes.

M_E : expiry-aware audit arm. The expiry-aware method runs in parallel with the primary method. It estimates an expected end date E_i only from explicit end-date or duration evidence. It does not assume a four-year duration. Candidate timing is

$$R_{ij} = C_j - E_i.$$

Very early candidates require stronger text evidence and CPV continuity. This arm accepts 504 links, compared with 544 under the main method. It is retained as an audit/sensitivity check, not promoted, because observed market behaviour shows many declared successors can be published before expected expiry.

6 Survival Modeling

The accepted-link decision from M_B defines the survival event. For episode i , τ_i is the time from award date A_i to the accepted successor’s publication date if $Y_i = 1$, or to the study cutoff (2025-12-31) if $Y_i = 0$ (administrative right-censoring). $Y_i = 0$ means no accepted observable successor was found before the cutoff; it is not proof of abandonment.

Kaplan–Meier. The non-parametric survivor function $S(t) = P(T > t)$ is estimated by [Kaplan and Meier \(1958\)](#)’s product-limit estimator, stratified by CPV segment, buyer type, and other covariates of interest. Group differences use the log-rank test.

Cox proportional hazards. The semi-parametric model is

$$h(t | X) = h_0(t) \exp(\beta_1 X_1 + \dots + \beta_p X_p),$$

with covariates selected for substantive relevance and data quality rather than automated search: CPV digital segment, buyer region, framework-agreement flag, validated-SIREN availability, and centered award year. The rule of one covariate per ten observed events ([Van Belle et al., 2002](#)) is respected with 544 events supporting 8 covariates. The proportional-hazards assumption is tested with Schoenfeld residuals ([Grambsch and Therneau, 1994](#)); violations are reported and interpreted descriptively rather than silently dropped or used to discard the model.

Parametric models. Exponential, Weibull, log-logistic, log-normal, and generalized-gamma models are compared by log-likelihood, AIC, and BIC, and checked graphically against the Kaplan–Meier curve, because parametric models allow probability extrapolation (e.g. 12/24-month successor probability) that the semi-parametric Cox model does not directly provide.

Temporal validation. The model is fit on episodes awarded 2015–2021 and evaluated on 2022–2025, reporting Harrell’s concordance index C on each split to assess discrimination and out-of-time stability, not to target a specific value.

Linkage sensitivity. Because the event is linkage-conditioned, the same Kaplan–Meier and Cox analyses are repeated under the strict ($M_B@0.80$), main ($M_B@0.70$), looser ($M_B@0.60$), and high-recall contrast ($M_C@0.70$) event definitions. A conclusion is reported as robust only when it is stable in sign and approximate magnitude across these arms.

7 Trend And Change-Point Detection

Quarterly awarded-episode counts $N_{s,q}$ are built for the overall cohort and each CPV digital segment from 2015Q2 (the first complete quarter) through 2025Q4, including zero-count quarters.

Change-point detection (PELT). Killick, Fearnhead and Eckley (2012)’s PELT algorithm minimizes a penalized segmentation objective on the z-standardized series with penalty $\beta = \lambda \log(n)$; the central result uses $\lambda = 1$ with sensitivity at 0.5 and 2.0, and a break is called stable only if it lies within one quarter under all three penalties.

Stationarity. Augmented Dickey–Fuller (unit-root null) and KPSS (level-stationary null) tests are run on each segment’s series. The two tests have opposite null hypotheses and are read together, not individually.

Regime detection (HMM). A 3-state Gaussian hidden Markov model is fit on the quarter-over-quarter *change* in episode count – not the level – for the overall cohort and the two highest-volume CPV segments, so states describe typical period-over-period direction (decline, plateau, growth) rather than absolute activity level. The Viterbi-decoded state sequence gives the current-quarter regime and its posterior probability. This complements PELT (which finds discrete historical breaks) with a continuously updated regime read; the two are not required to agree, and disagreement is reported honestly rather than reconciled.

Recent direction. An OLS slope over the latest 12 quarters gives an `increasing/decreasing/stable_or_uncertain` label at exploratory $\alpha = 0.10$, uncorrected for multiple testing. None of these methods forecasts future values or identifies the cause of a break; causal attribution requires documentary or stakeholder evidence not available in BOAMP alone. Monetary and duration trend series are omitted because no canonical awarded-amount field is validated at episode grain, and 2025’s duration-field completeness jump is a measurement change, not a genuine shift in contract durations.

8 Technology Segmentation

The internship guide’s L2 deliverable specifies a supervised technology taxonomy: 300–500 manually annotated contracts across 8–12 classes, a TF–IDF+logistic-regression/SVM baseline evaluated by macro-F1 and confusion matrix, and an optional CamemBERT comparison. **This was not built.** Two independent annotators with a Cohen’s kappa agreement statistic are required for a defensible L2 corpus, and no second qualified annotator was available within this session’s scope; producing single-pass AI-assisted labels and calling their agreement "kappa" would repeat exactly the self-consistency problem this project has documented for the successor-linkage benchmark (§4) instead of avoiding it.

Every technology segment referenced in this report – cohort selection, Cox covariates, quarterly trend series – therefore uses CPV divisions 32 (telecommunications equipment), 35 (security), 48 (software), and 72 (IT services) as a reproducible substitute. This is a real scope reduction, not a disguised classifier: CPV divisions are official, zero-missingness EU categories under Regulation 213/2008, but they are coarser than a learned taxonomy and cannot distinguish sub-themes such as cloud versus on-premise infrastructure within a division. A future session with a second qualified annotator could complete L2 using the same blinded-sample-plus-adjudication scaffolding already built for linkage review (`scripts/prepare_independent_link_review.py`, `scripts/ingest_annotations.py`, `scripts/adjudicate_annotations.py`).

9 Dev Results

Method	Threshold	Accepted	Precision	Recall	FPR	Coverage
M_A_deterministic	70.000	10	0.600	0.128	0.035	0.075
M_B_text_ranking	70.000	24	0.875	0.447	0.035	0.179
M_C_weighted_gated	70.000	47	0.596	0.596	0.126	0.351
M_D_fellegi_sunter	65.000	4	0.000	0.000	0.023	0.030

Table 1: Unweighted dev metrics on all labelled current benchmark frames.

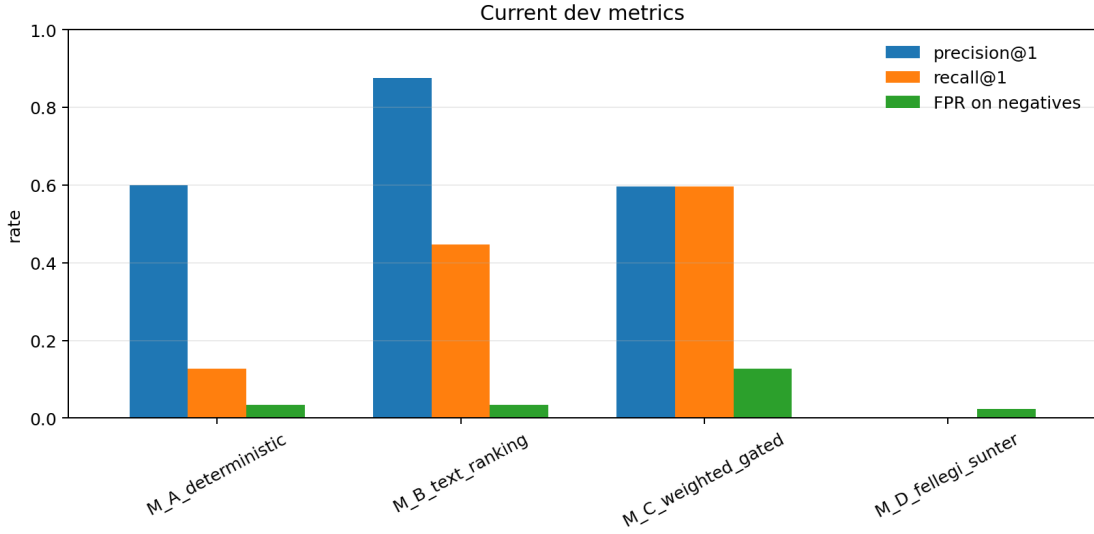


Figure 1: Dev split method comparison generated from the current benchmark evaluation JSON.

10 Internal Held-Out Reference Results

The table below is an internal method-comparison diagnostic. It must not be presented as external validation. The primary comparison is anchor-level exact-successor performance: each anchor can produce one accepted successor or abstain.

$$\text{precision} = \frac{TP}{TP + FP}, \quad \text{recall} = \frac{TP}{TP + FN}, \quad \text{FPR} = \frac{FP}{FP + TN}.$$

For this project, false-positive control is prioritised because false links fabricate survival events.

Method	Threshold	Accepted	Precision	Recall	FPR	Coverage
M_A_deterministic	70.000	6	0.167	0.045	0.105	0.100
M_B_text_ranking	70.000	5	0.800	0.182	0.000	0.083
M_C_weighted_gated	70.000	15	0.600	0.409	0.105	0.250
M_D_fellegi_sunter	65.000	1	0.000	0.000	0.026	0.017

Table 2: Unweighted held-out metrics on all bootstrap-labelled reference frames.

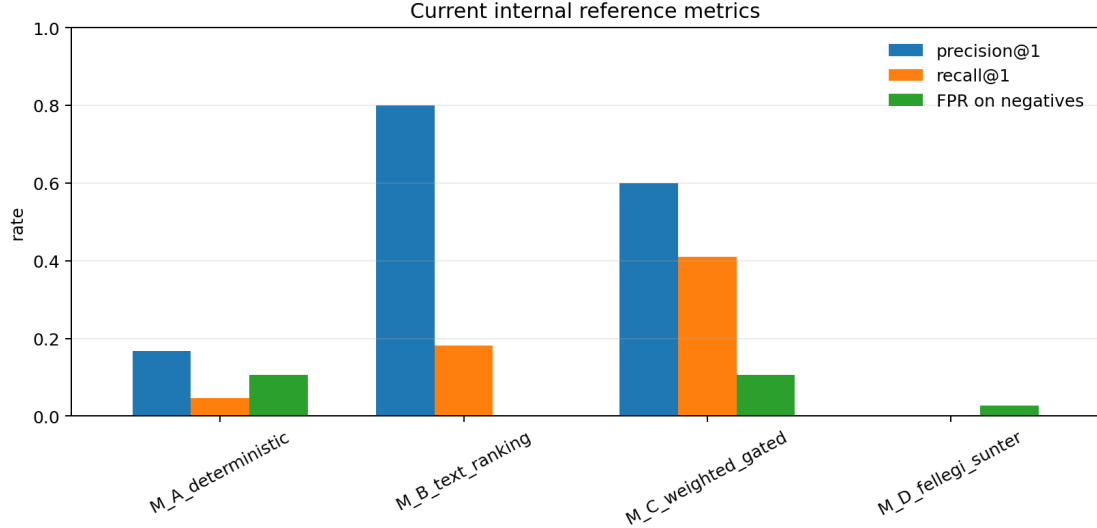


Figure 2: Held-out internal method comparison generated from the development-reference JSON.

11 Quality Evidence And Interpretation

The held-out internal result supports retaining a conservative baseline for continued work; it does not establish final accuracy. M_B has only 5 accepted held-out links, so its precision estimate is necessarily sample-sensitive; one changed decision would move the number materially. Within this bootstrap reference, the direction is coherent: M_B gives the best precision and the lowest false-positive rate among useful methods, while M_C shows the expected precision-recall trade-off.

The pair-level ROC and precision-recall curves should be read as score-ranking diagnostics, not as the final operating decision. The final pipeline decision is anchor-level: choose one successor or abstain. The ROC curve is stepped rather than smooth because the held-out reference contains a finite number of labelled pairs and many tied or near-tied scores.

The emphasis on precision-recall for rare positive decisions is supported by [Davis and Goadrich \(2006\)](#) and [Saito and Rehmsmeier \(2015\)](#). These sources justify the diagnostic choice; they do not validate the bootstrap labels, numerical results, or selected threshold. Generic web illustrations are therefore treated as presentation aids only, not as academic evidence.

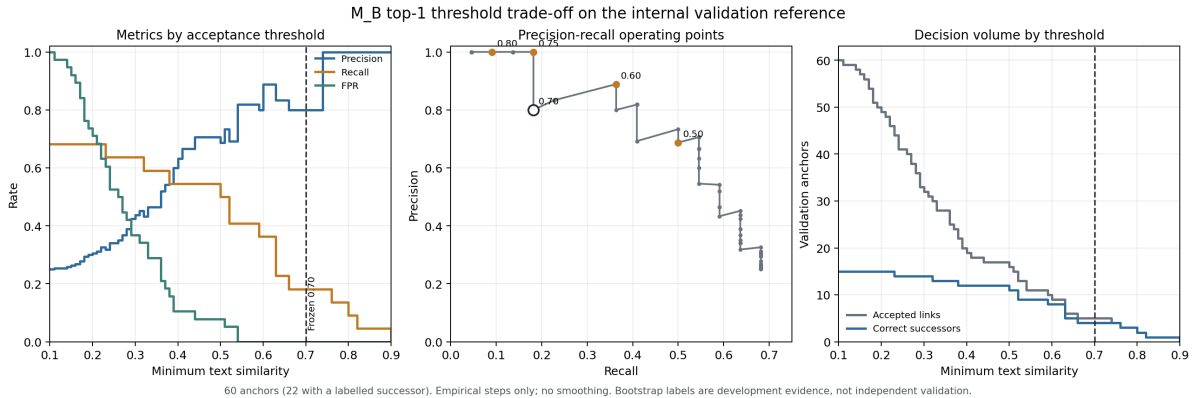


Figure 3: Project-specific, unsmoothed anchor-level threshold trade-off for M_B. The figure is empirical evidence from the internal validation reference, not an idealised illustration.

At threshold 0.60, this validation sample contains 8 correct successors among 9 accepted links, compared with 4 among 5 at 0.70. Thus 0.60 performs better on this particular sample.

Development evidence points in the opposite direction: precision is 0.800 and FPR is 0.058 at 0.60, compared with precision 0.875 and FPR 0.035 at 0.70. The lower threshold is therefore reported as a sensitivity arm rather than promoted after inspecting validation.

12 Model-Assisted Challenge Review

A separate blinded challenge review sampled 20 current accepted links, 20 high-similarity structural-negative candidates, and 20 buyer-declared relationships. Its provenance is model-assisted rather than independent human specialist review. Among the 20 accepted links, 14 were confirmed, 5 were rejected, and 1 was uncertain. Conservative reviewed precision was 0.700, with exact 95% interval $[0.457, 0.881]$. This diagnostic falls below the 0.80 point target and exposes useful failure modes. It does not establish independent human validation, but it is sufficient to justify keeping the current claim narrow and the lower threshold unpromoted.

13 Modeling Tables

The modeling-ready tables include strict, primary, broad, and non-match target columns, plus observable features. They now include the Fellegi–Sunter columns `fs_match_weight` and `fs_match_probability`, so modeling and evaluation use the same feature state.

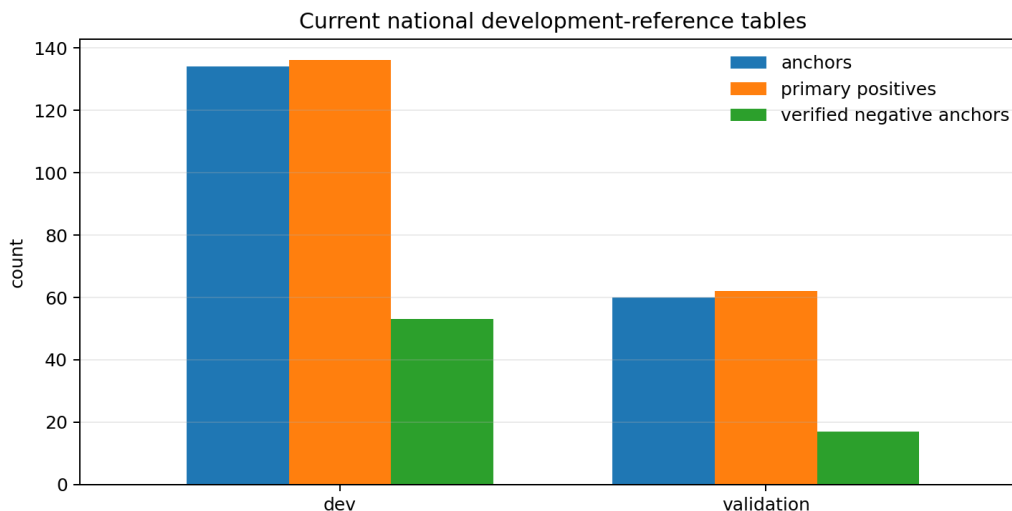


Figure 4: Current benchmark modeling table sizes and label support.

14 Survival Results

The survival dataset uses the main accepted links as events and all other episodes as right-censored observations at the study cutoff. The main variant contains 544 events and 3,256 censored observations (14.316% event rate). Segment event rates are 0.132 for CPV-32, 0.204 for CPV-35, 0.117 for CPV-48, and 0.143 for CPV-72. CPV-35 has the highest observed successor rate in the main arm.

Kaplan–Meier. The estimated probability of an observable successor is 4.621% by 12 months, 6.733% by 24 months, and 17.535% by 60 months. The Kaplan–Meier median survival time is **not reached**: the curve never falls below 0.5 within the observation window, so no median is reported. This is distinct from the 31.82-month median delay *among linked events only*, which

conditions on the event having occurred. A multivariate log-rank test across CPV segments gives statistic 23.45 ($p = 3.26e - 05$).

Cox model. The parsimonious model (8 covariates, 544 events) gives:

Covariate	HR	95% CI	p
framework_flag	1.751	[1.435, 2.136]	3.38e-08
has_validated_siren	1.082	[0.885, 1.323]	0.443
award_year_centered	1.107	[1.066, 1.149]	1.21e-07
digital_segment_CPV-35	1.553	[1.218, 1.981]	0.000383
digital_segment_CPV-48	0.828	[0.638, 1.073]	0.153
digital_segment_CPV-72	1.056	[0.850, 1.310]	0.624
buyer_region_Normandie	0.800	[0.640, 1.000]	0.0501
buyer_region_Pays de la Loire	1.003	[0.815, 1.234]	0.979

Table 3: Cox hazard ratios, main linkage arm. In-sample C-index 0.626.

Framework-agreement episodes and CPV-35 carry the largest hazard ratios among substantively interpretable covariates. These are descriptive associations with the observed hazard, not causal effects.

Covariate	Test statistic	p	PH assumption
award_year_centered	70.733	4.09e-17	violated
buyer_region_Normandie	0.608	0.436	ok
buyer_region_Pays de la Loire	2.455	0.117	ok
digital_segment_CPV-35	0.041	0.84	ok
digital_segment_CPV-48	0.671	0.413	ok
digital_segment_CPV-72	3.723	0.0537	ok
framework_flag	7.040	0.00797	violated
has_validated_siren	6.763	0.00931	violated

Table 4: Schoenfeld-residual proportional-hazards tests.

Proportional-hazards diagnostic. The assumption is rejected ($p < 0.05$) for **award_year_centered**, **framework_flag**, **has_validated_siren**. These coefficients are therefore reported as time-averaged associations rather than constant hazard ratios; stratification or time-interaction terms would be the next refinement if individualized prediction were required.

Temporal validation. Training on 2015-2021 (2,470 episodes, 392 events) and evaluating on 2022-2025 (1,330 episodes, 152 events) gives C-index 0.607 in-sample versus 0.518 out-of-time. This gap indicates the model is not validated for individualized operational prediction; it is retained as a descriptive risk-factor summary.

Model	Parameters	Log-likelihood	AIC	BIC
GeneralizedGamma	3	-3757.5	7521.0	7539.7
LogNormal	2	-3784.6	7573.2	7585.7
LogLogistic	2	-3807.3	7618.6	7631.1
Weibull	2	-3814.7	7633.4	7645.8
Exponential	1	-3829.7	7661.3	7667.6

Table 5: Parametric survival model comparison.

Parametric models. `GeneralizedGamma` has the lowest AIC and BIC and was checked graphically against the Kaplan–Meier curve before selection; it is used for the exported 12/24-month conditional successor probabilities (`survival_conditional_probabilities.csv`), each with a 500-draw episode-bootstrap interval. These are estimated probabilities of an *identifiable observable successor procurement*, not certified renewal probabilities, and no active Gigalis portfolio was available to score, so predictions cover the study cohort rather than a live deployment set.

Detectability and selection diagnostic. Comparing linked and censored episodes on standardized mean differences, the largest gap is `text_length_chars` (SMD 0.262). This indicates possible differential detectability across episode characteristics, not proof of causal linkage bias; it cannot be fully separated from genuine heterogeneity in renewal behaviour using BOAMP alone.

Linkage sensitivity. Event counts range from 296 to 1,332 across the four retained linkage arms, so absolute probabilities are linkage-sensitive by construction. Cox effects under the same four arms:

Covariate	Strict	Main	Looser	High-recall	Robustness
<code>award_year_centered</code>	1.017	1.107	1.136	1.101	Robust
<code>buyer_region_Normandie</code>	0.803	0.800	0.794	0.726	Mostly Robust
<code>buyer_region_Pays de la Loire</code>	0.977	1.003	1.023	0.891	Linkage Sensitive
<code>digital_segment_CPV-35</code>	1.792	1.553	1.488	1.484	Robust
<code>digital_segment_CPV-48</code>	0.652	0.828	0.807	0.885	Mostly Robust
<code>digital_segment_CPV-72</code>	1.108	1.056	1.036	1.557	Linkage Sensitive
<code>framework_flag</code>	1.281	1.751	1.953	1.623	Robust
<code>has_validated_siren</code>	1.193	1.082	1.107	1.650	Linkage Sensitive

Table 6: Cox hazard ratios across linkage-definition sensitivity arms.

Framework flag, CPV-35, and centered award year are the most robust associations; buyer-region and CPV-72 effects are linkage-sensitive and should not be over-interpreted.

15 Trend Results

Segment	Direction	Slope/qtr	p	Last stable PELT break	HMM regime
Overall	<code>stable_or_uncertain</code>	-0.11	0.921	–	growth
CPV-32	<code>stable_or_uncertain</code>	-0.01	0.989	2020Q2	growth
CPV-35	<code>stable_or_uncertain</code>	0.03	0.923	–	–
CPV-48	decreasing	-0.84	0.032	2024Q1	–
CPV-72	<code>stable_or_uncertain</code>	0.70	0.285	2021Q1	growth

Table 7: Current trend signal matrix: OLS 12-quarter slope, PELT breaks, and HMM current regime (Overall and top-2 segments only).

CPV-48 is the only segment with a statistically distinguishable 12-quarter decline at the exploratory $\alpha = 0.10$ level; the rest are `stable_or_uncertain` by this signal.

Segment	ADF (H0: unit root)	KPSS (H0: level stationary)	Note
Overall	stat=-5.890, p=0.000	stat=0.268, p=0.100	
CPV-32	stat=-6.086, p=0.000	stat=0.808, p=0.010	
CPV-35	stat=-1.684, p=0.439	stat=0.444, p=0.058	
CPV-48	stat=-1.749, p=0.406	stat=0.545, p=0.032	
CPV-72	stat=-5.439, p=0.000	stat=0.611, p=0.022	

Table 8: Stationarity diagnostics per segment.

ADF and KPSS test opposite null hypotheses and are read jointly; disagreement between them means the series is not cleanly classified as stationary or non-stationary over this 43-quarter window.

The HMM regime column above reports each covered segment’s current-quarter regime and posterior probability, fit on the quarter-over-quarter change in episode count rather than its level. Because the model is fit on noisy, low-count quarterly data, the **plateau** state is a data-driven middle tier rather than a change centered exactly at zero, and it is read alongside **decline/growth** rather than as "no change." The HMM’s current-quarter regime and the 12-quarter OLS slope are complementary diagnostics computed over different windows; they are not required to agree, and this report does not adjust either to match the other.

16 Defensible Decision

The final project decision is to keep **M_B_text_ranking @ 0.70** as the frozen conservative observable-successor baseline and use the stricter, looser, weighted-gated, and expiry-aware variants as required sensitivity analyses. The threshold is not claimed to be optimal: 0.60 performs better on the small bootstrap validation split, while 0.70 has better precision and false-positive control on development evidence. Moreover, the completed model-assisted review confirmed only 14 of 20 sampled production links at 0.70 conservatively, so lowering an unreviewed threshold would not support a stronger accuracy claim. This is a precision-first design. Low recall is accepted as an explicit trade-off, but the observed event rate is not a mathematical lower bound: missed successors push it downward while residual false links can push it upward. It is therefore a linkage-conditioned indicator whose absolute value must be read with the strict, looser, weighted-gated, and expiry-aware sensitivity results. Independent specialist review is required only before claiming externally validated precision or promoting a new threshold, not to complete this linkage-conditioned descriptive study. If that review shows weaker precision, the claim should be narrowed further rather than forcing a more complex model.

17 Limitations And Robustness

- BOAMP does not consistently encode legal renewal status; accepted links are observable successor procurements, not legal renewal proof.
- The current reference labels come from deterministic bootstrap rules; pass agreement is self-consistency, not independent human agreement.
- The held-out reference is small: 22 positive anchors and 5 accepted **M_B** links. The selected threshold should not be over-tuned and its accuracy remains provisional.
- Buyer standardisation is improved but remains an important risk area. The legal-form audit is retained to catch name-only and cross-legal-form cases.
- Missing duration and expiry fields are not imputed. This avoids creating false timing certainty.

- The 3-state HMM regime labels are fit on noisy, short (43-quarter) count series; state-count and window choices are not exhaustively validated, and a ‘plateau’ label is a relative middle tier rather than a change centered at zero.
- Technology segmentation uses CPV divisions, not a learned taxonomy (§8); this is a real scope reduction, disclosed rather than concealed.

18 Perspectives

18.1 Causal Inference (Outline Only)

The internship guide frames a distinct causal question, separate from this report’s descriptive linkage/survival/trend results: does a Gigalis pooled framework agreement actually change member purchasing behaviour (amounts, frequency, segments), or does Gigalis simply capture demand that would have existed anyway? **This question is not answered here, and no causal estimate is computed in this repository.** Answering it requires data this project’s BOAMP-only corpus does not contain: which buyers are Gigalis members, and when each joined. Attempting a substitute causal claim from BOAMP alone – for example treating CPV-segment membership or region as a proxy treatment – would confound procurement domain and geography with Gigalis adoption, and this project has consistently avoided exactly this kind of unsupported inferential leap elsewhere (survival hazard ratios are reported as associations, not effects, for the same reason).

If Gigalis-internal membership and adoption-date data were joined to this corpus, the natural identification strategy is a **staggered-adoption difference-in-differences** design ([Callaway and Sant’Anna, 2021](#)), because members join the central purchasing body at different calendar dates rather than simultaneously – the specific setting that motivates that estimator over classical two-period DiD. The parallel-trends assumption (comparable purchasing trajectories absent Gigalis membership) would need to be assessed with pre-adoption trend plots per member cohort. Propensity-score matching ([Angrist and Pischke, 2009](#)) is a secondary candidate if adoption is not sharply timed but depends on observable buyer characteristics (size, budget, sector); regression discontinuity would apply only if a hard eligibility threshold (e.g. a population cutoff) determines Gigalis access, which is not currently known to be the case. Any of these designs should state its identifying assumptions as an explicit causal graph ([Pearl, Glymour and Jewell, 2016](#)) before estimation, not after.

This section is a design outline for future work conditional on Gigalis supplying membership data, not a promise that such data will become available within this internship.

19 Recommended Reporting Position

The technical report should defend the project as a reproducible measurement pipeline, not as a black-box prediction system. The strongest statement is:

Because legal renewal status is not directly observed in BOAMP, the study estimates observable successor procurements. Candidate generation restricts comparison to the same buyer and a plausible future interval; the selected method then accepts only the strongest textually similar candidate above a fixed threshold. This prioritises precision over recall, which is appropriate because false positive links would create artificial survival events.

20 Current Source Files

- `data/processed/boamp/linkage_evaluation_dev.json`

- `data/processed/boamp/linkage_evaluation_validation.json`
- `data/processed/boamp/benchmark/modeling/modeling_summary.json`
- `data/processed/boamp/benchmark/benchmark_manifest.json`
- `data/processed/boamp/survival_dataset_summary.json`
- `data/processed/boamp/linkage_candidates_summary.json`
- `data/processed/boamp/survival_analysis_summary.json`
- `data/processed/boamp/survival_cox_results.csv`
- `data/processed/boamp/survival_ph_diagnostics.csv`
- `data/processed/boamp/survival_parametric_comparison.csv`
- `data/processed/boamp/survival_cox_linkage_sensitivity.csv`
- `data/processed/boamp/trend_analysis_summary.json`
- `data/processed/boamp/trend_signal_matrix.csv`
- `DATA_QUALITY_REPORT.md`
- `TREND_ANALYSIS_REPORT.md`
- `SURVIVAL_ANALYSIS_REPORT.md`
- `PROJECT_WORK_PROTOCOL.md`
- `REVIEW_AUDIT_RESULTS.md`
- `EXECUTIVE_SUMMARY.md`

21 Methodological References

The source and identity definitions follow the official BOAMP and INSEE documentation. CPV interpretation follows Commission Regulation (EC) No 213/2008. TF-IDF cosine similarity follows the standard vector-space definition. Record linkage, survival estimation, proportional-hazards diagnostics, and change-point detection are supported by the original Fellegi–Sunter (1969), Davis–Goadrich (2006), Saito–Rehmsmeier (2015), Kaplan–Meier (1958), Cox (1972), Grambsch–Therneau (1994), and PELT (Killick et al., 2012) methods. Regime detection follows the Markov-switching framework of Hamilton (1989). The causal-inference outline in §18.1 draws on Angrist and Pischke (2009), Pearl, Glymour and Jewell (2016), Athey and Imbens (2017), and Callaway and Sant’Anna (2021). None of these sources validates this project’s own labels, numerical results, or thresholds; they justify method choices only. Full URLs and the specific design implications are recorded in `METHODOLOGICAL_REFERENCES.md`.

- [Official BOAMP API \(DILA\)](#).
- [INSEE SIREN definition](#) and [INSEE SIRET definition](#).
- [Commission Regulation \(EC\) No 213/2008 on CPV](#) and [Directive 2014/24/EU, Article 33](#).
- [Cosine similarity documentation](#).

- [Davis–Goadrich \(2006\)](#) and [Saito–Rehmsmeier \(2015\)](#) on ROC and precision-recall evaluation for skewed binary decisions.
- [Fellegi–Sunter \(1969\)](#), [Kaplan–Meier \(1958\)](#), and [Cox \(1972\)](#).
- [Grambsch–Therneau \(1994\)](#) and [Killick et al. \(2012\)](#).
- [Hamilton \(1989\)](#) on Markov-switching regime models, the framework underlying the trend HMM.
- [Angrist and Pischke \(2009\)](#), [Pearl, Glymour and Jewell \(2016\)](#), [Athey and Imbens \(2017\)](#), and [Callaway and Sant’Anna \(2021\)](#) on causal identification, cited only to support the outlined-not-executed causal-inference perspective.